

Econ 139: Human Capital Part 2

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The Causal Impact of Education

- Last class: individuals (and countries) with higher levels of education also have higher income
- But this is not causal evidence: it could be some other factor that varies between individuals that generates differences in income
- These sets of slides focus on various research designs that have been proposed to study this question

The Causal Impact of Education

- We will go through a few different types of research designs
 - Twin studies
 - Compulsory schooling laws
 - **Regression discontinuity designs**

Twin Studies

- There is considerable work in social sciences and biology that makes use of twin studies
- Basic idea: we are usually concerned about “other differences” between individuals with different levels of schooling driving difference
- But with twins, these differences are more limited: in the case of identical twins they have the same genetics



Twin Studies

- Ashenfelter and Rouse (1998) surveyed identical twins at Twinsburg Twins Festival
- Found an additional year of schooling is associated with about **9 percent**
- Smaller than what we found in the ACS, though not widely different. Consistent with lots of evidence at the time



Compulsory Schooling Laws

- In the U.S. states have compulsory schooling laws which dictate at what age a student can legally drop out of school (usually 16 or 18)
- Additionally, when you start school depends exactly where your birthday falls in the Calendar year
- Angrist and Krueger (1991) realized this could be used to identify the causal impact of an additional year of schooling for individuals on the margin of dropping out

Compulsory Schooling Laws

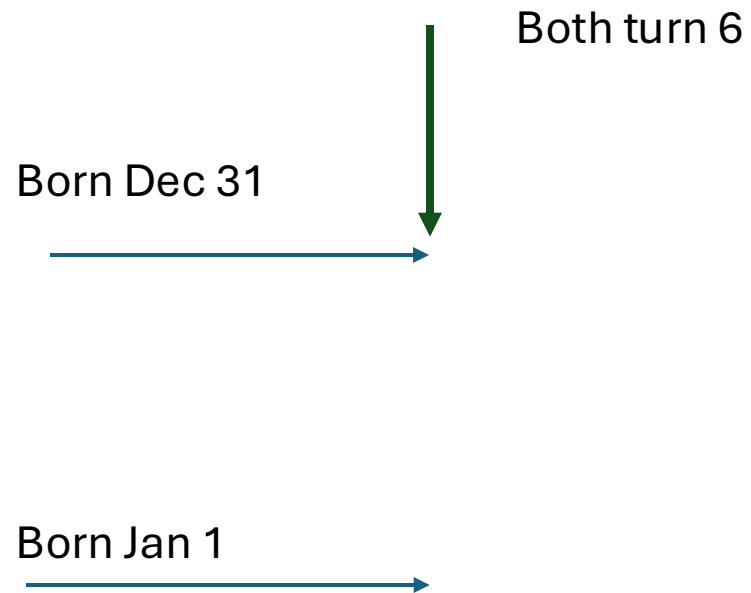
Born Dec 31



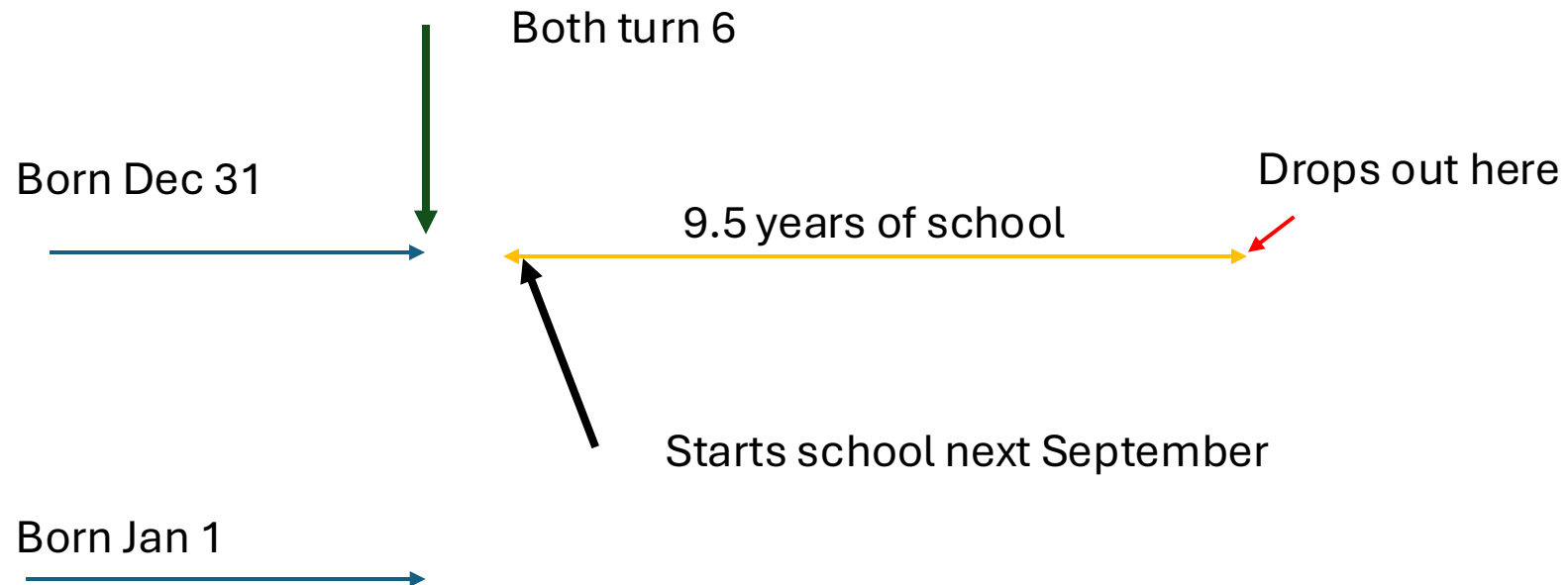
Born Jan 1



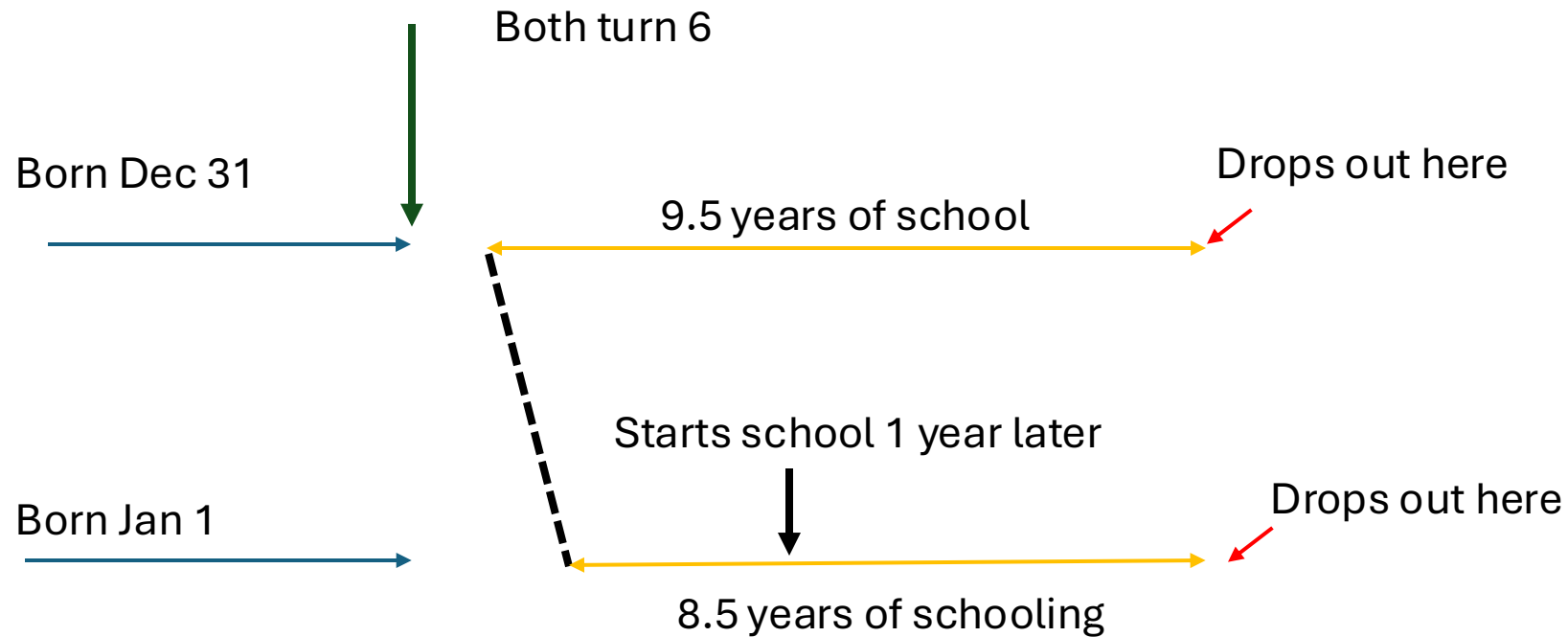
Compulsory Schooling Laws



Compulsory Schooling Laws



Compulsory Schooling Laws



Quarter of Birth

- Angrist and Kruger (1991) find a return to a year of schooling of 7.5 percent
- Overall, while there have been many clever studies using different populations, different sources of variation, a lot agree on the magnitude of the earnings impact
- **Summary:** Across many studies (not just the two so far) it is often found that an additional year of schooling has around a 7-10 percent **causal impact** on your overall earnings

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 - Treatment: you are interested in the effect of some policy on individuals
 - Ex: how does receiving financial aid impact your college choice

Regression Discontinuity Design

- Next, we are going to introduce another approach to estimating causal impacts which has proven very useful in the economics of education: the regression discontinuity design
- **Basic Setup:**
 - Treatment: you are interested in the effect of some policy on individuals
 - Ex: how does receiving financial aid impact your college choice
 - The treatment is determined by a cutoff rule
 - Ex: if you get a high enough test score you are eligible for financial aid

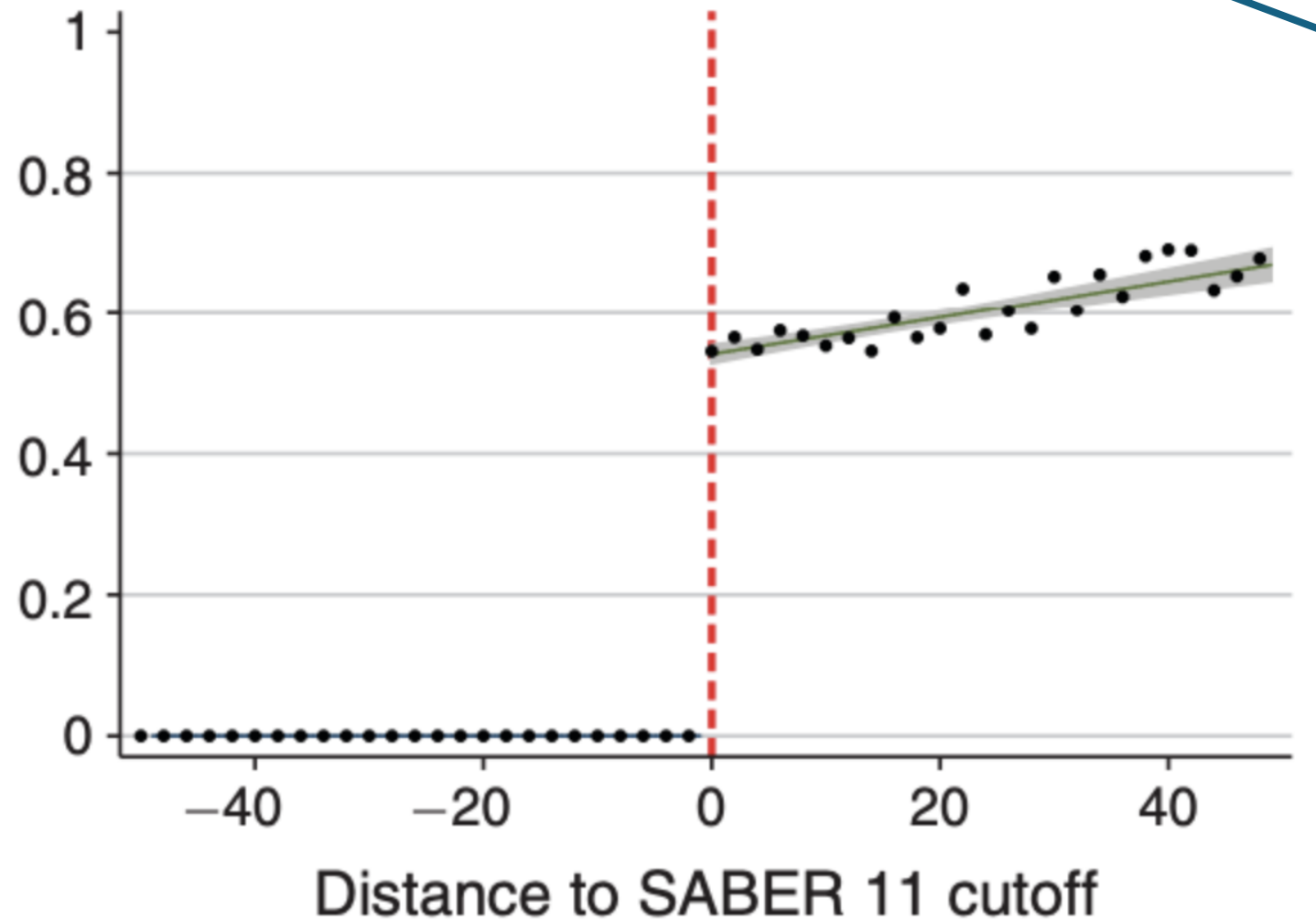
Regression Discontinuity Design

- The first use of this design comes from Campbell and Thistlewaite (1960), but popularity of design did not increase until 2000s
- In the original paper, they were interested in the impact of receiving a scholarship on student outcomes
- We will be studying a similar question analyzed in Londono-Velez et al. (2020)

Londono-Velez et al. (2020)

- This paper studies the impact of financial aid on students in Colombia
- In 2014, government launched a merit-based program (Ser Pilo Paga) for low-income students
 - Large program – covered full tuition for attendance at a High Quality Accredited School
- Key feature: eligibility depended on a cutoff rule. If you received a high enough score on a test you were eligible for the program

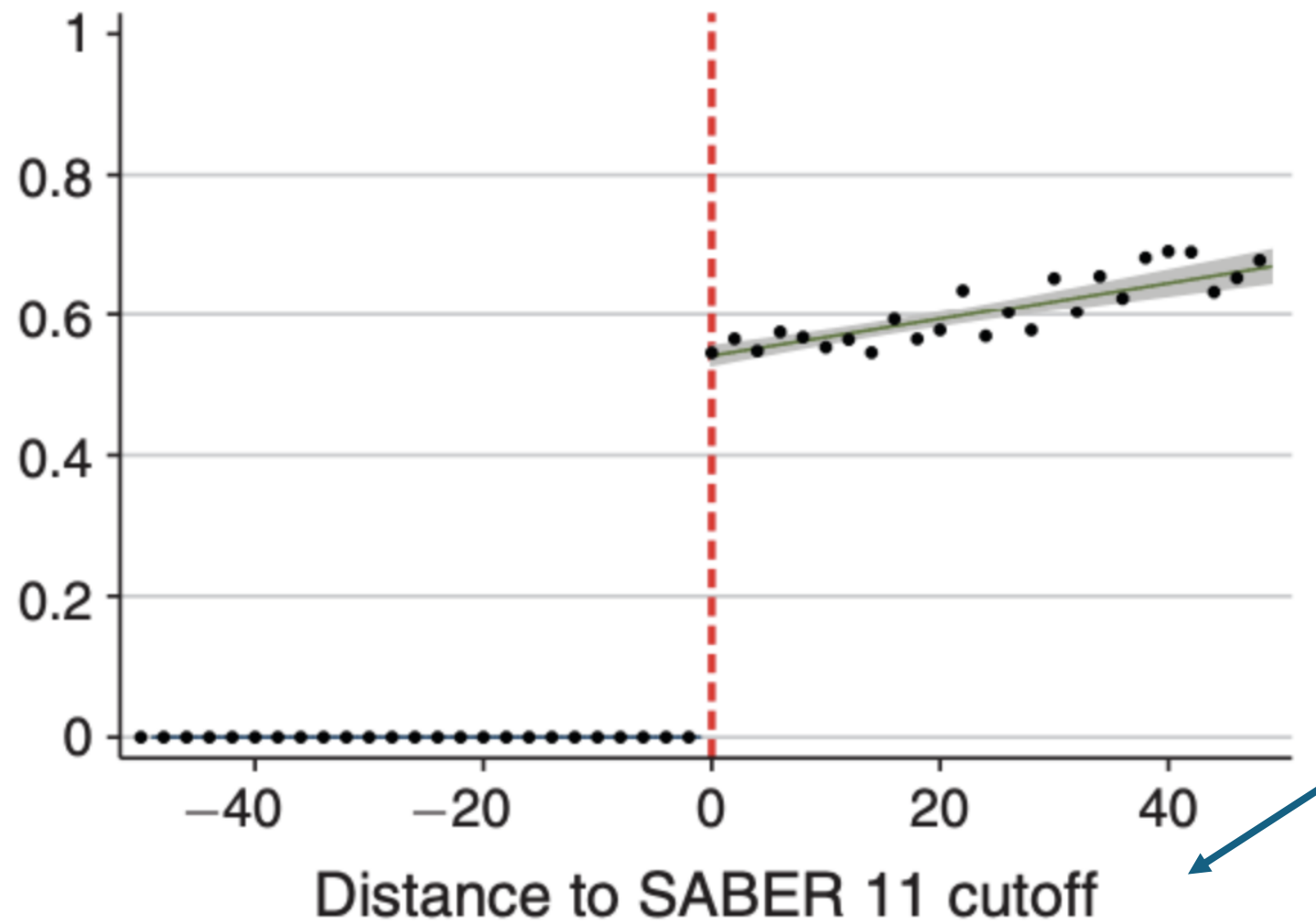
Panel A. $R_i = \text{SABER 11 test score}$



SABER 11 is a test that all high school students take when exiting high school

Sample restricted to SISBEN-eligible individuals

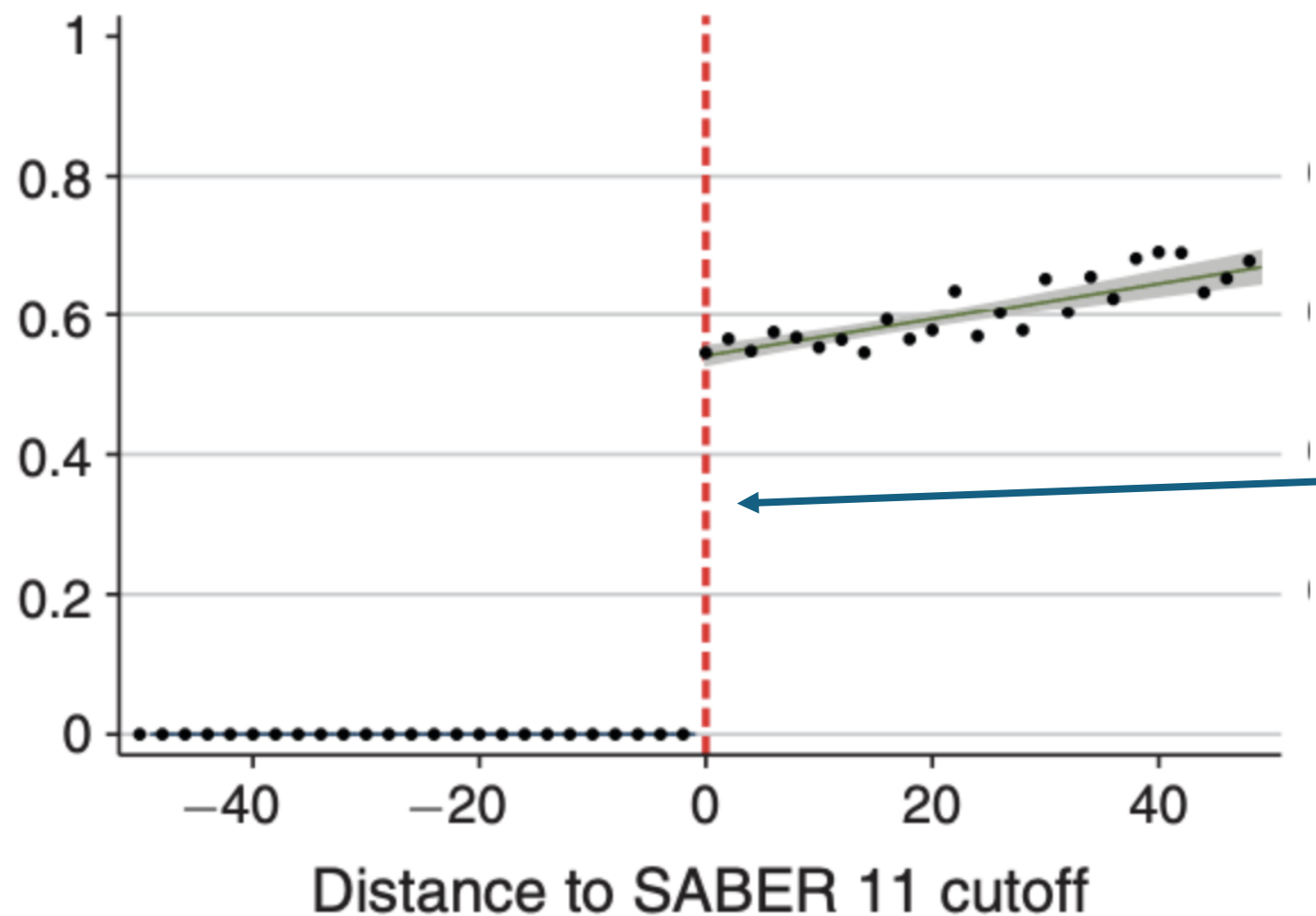
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Sample restricted to SISBEN-eligible individuals

This plot is restricted to eligible candidates which depends on socioeconomic status

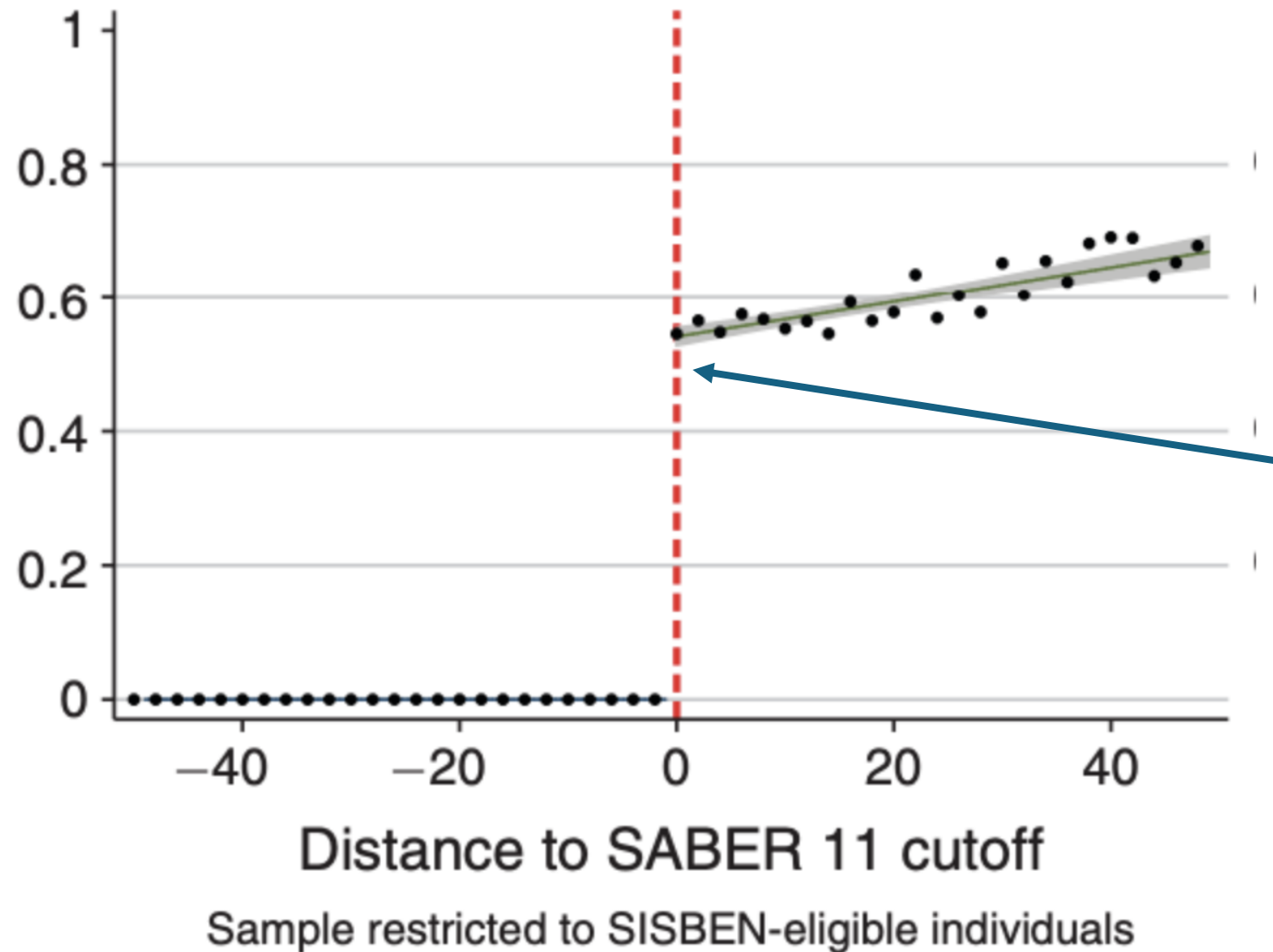
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Sample restricted to SISBEN-eligible individuals

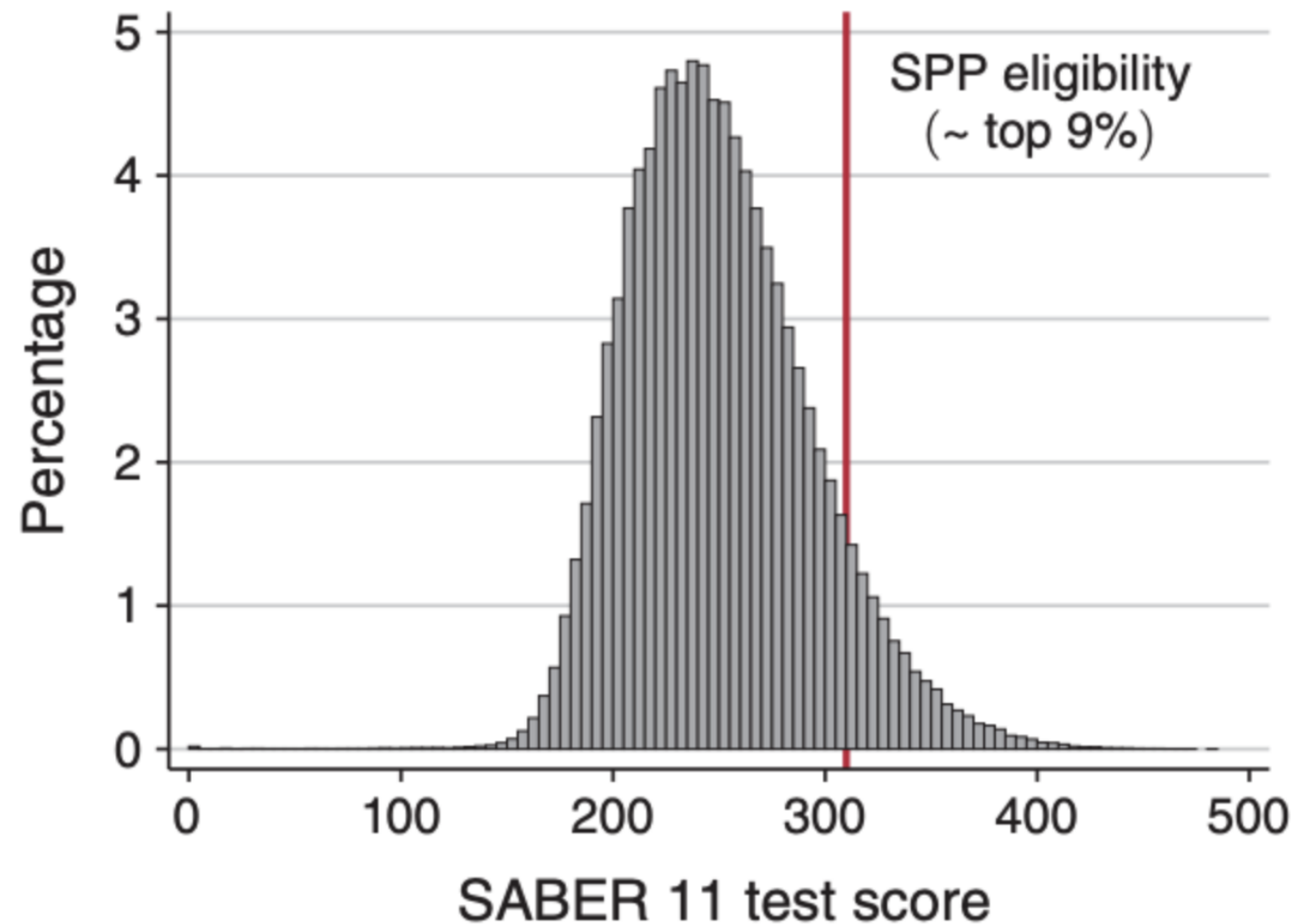
Outcome is probability of participating in program – about 55 percent of students eligible (at the cutoff) take up the program

Panel A. $R_i = \text{SABER 11 test score}$



Some students here either did not attend college or did not attend a high quality college, so did not take up the program. Because there is partial takeup, this is referred to as a fuzzy regression discontinuity

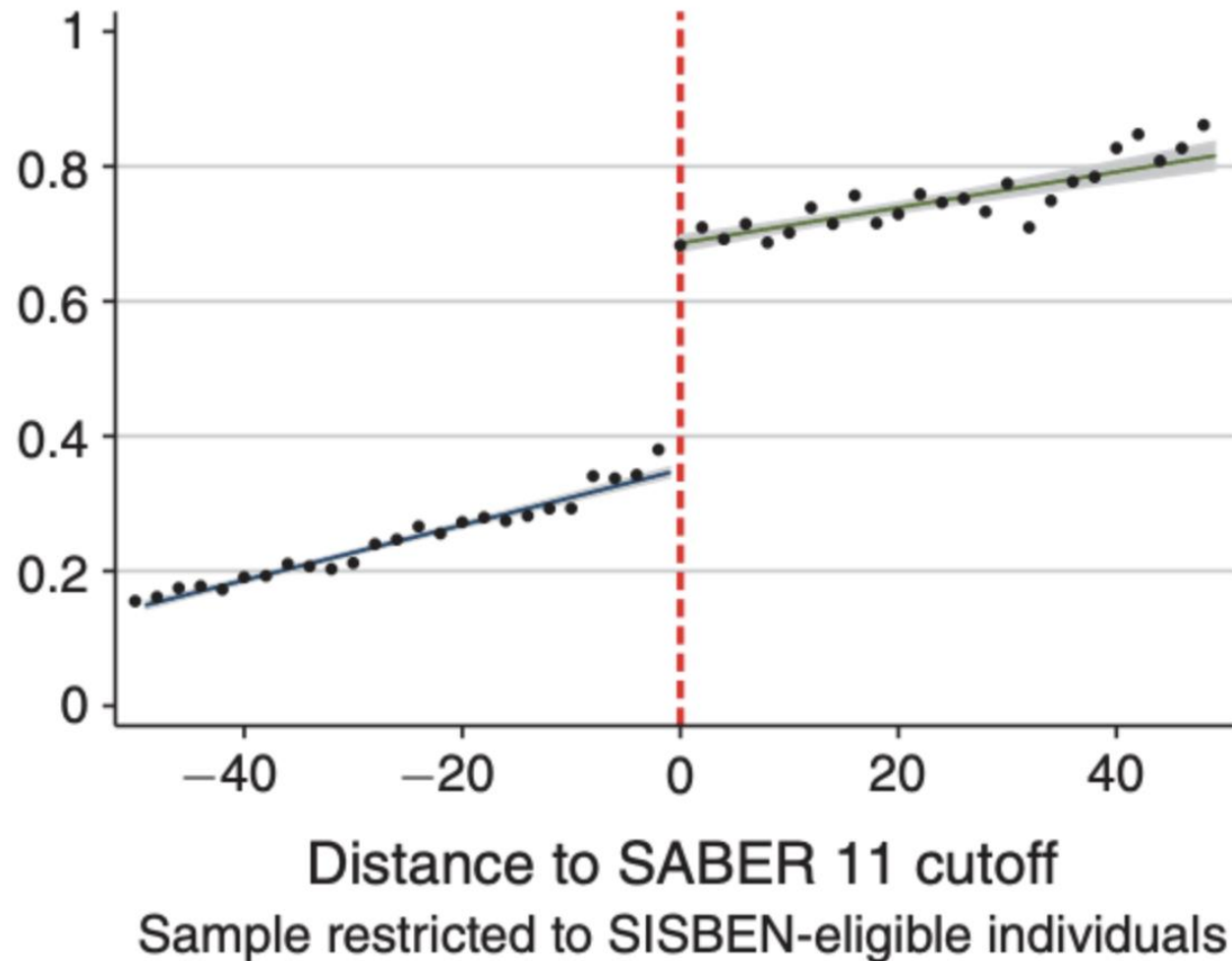
Panel A. Merit: SABER 11 test score $\geq 310/500$



Assumption of RD:
individuals cannot
precisely impact
their test score

If individuals could
manipulate the
score, we would
see a spike right at
the cutoff

Panel A. $R_i = \text{SABER 11 test score}$

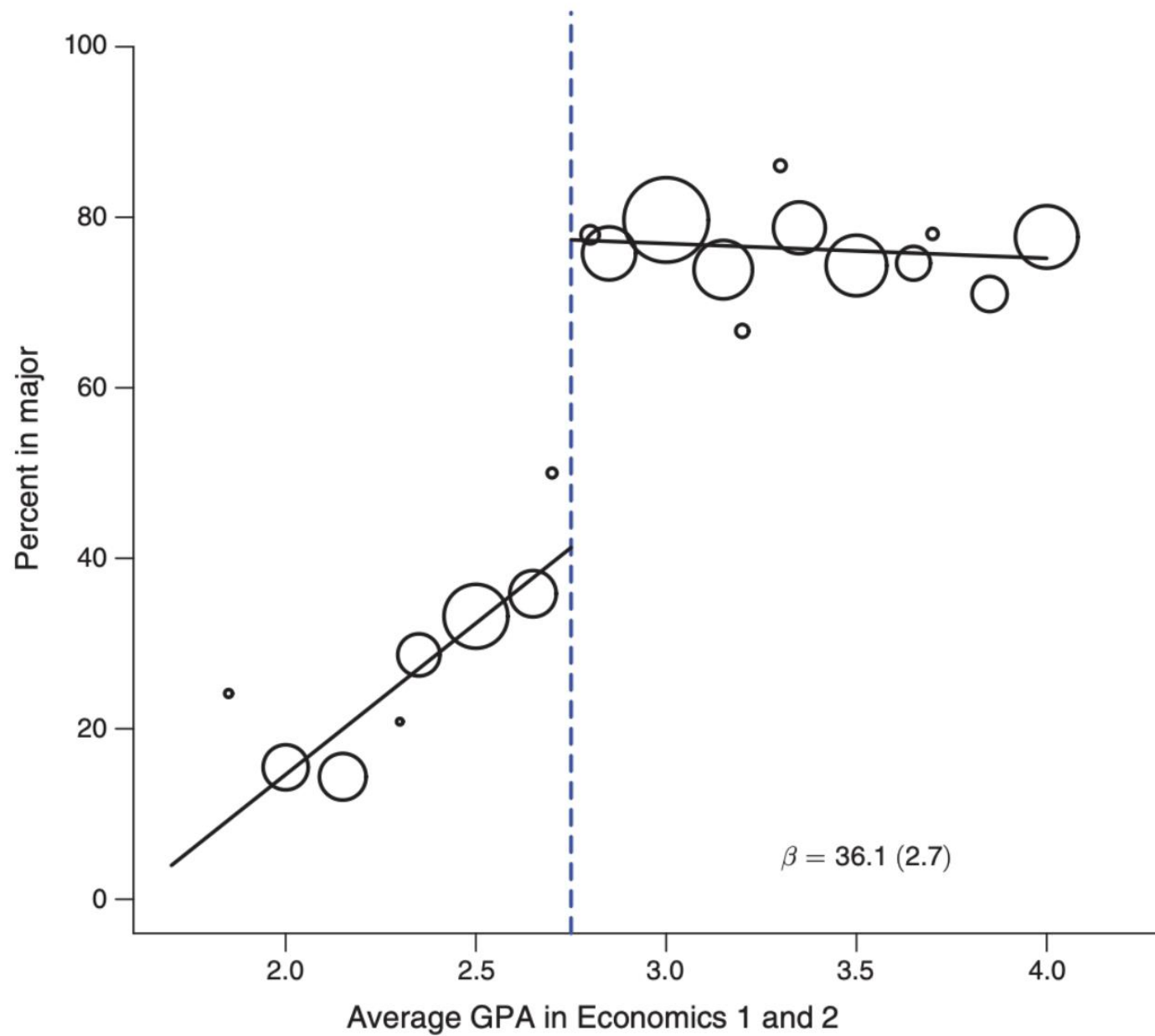


Availability of
financial aid has
huge impact on
college
attendance

About 32
percentage point
increase in college
attendance

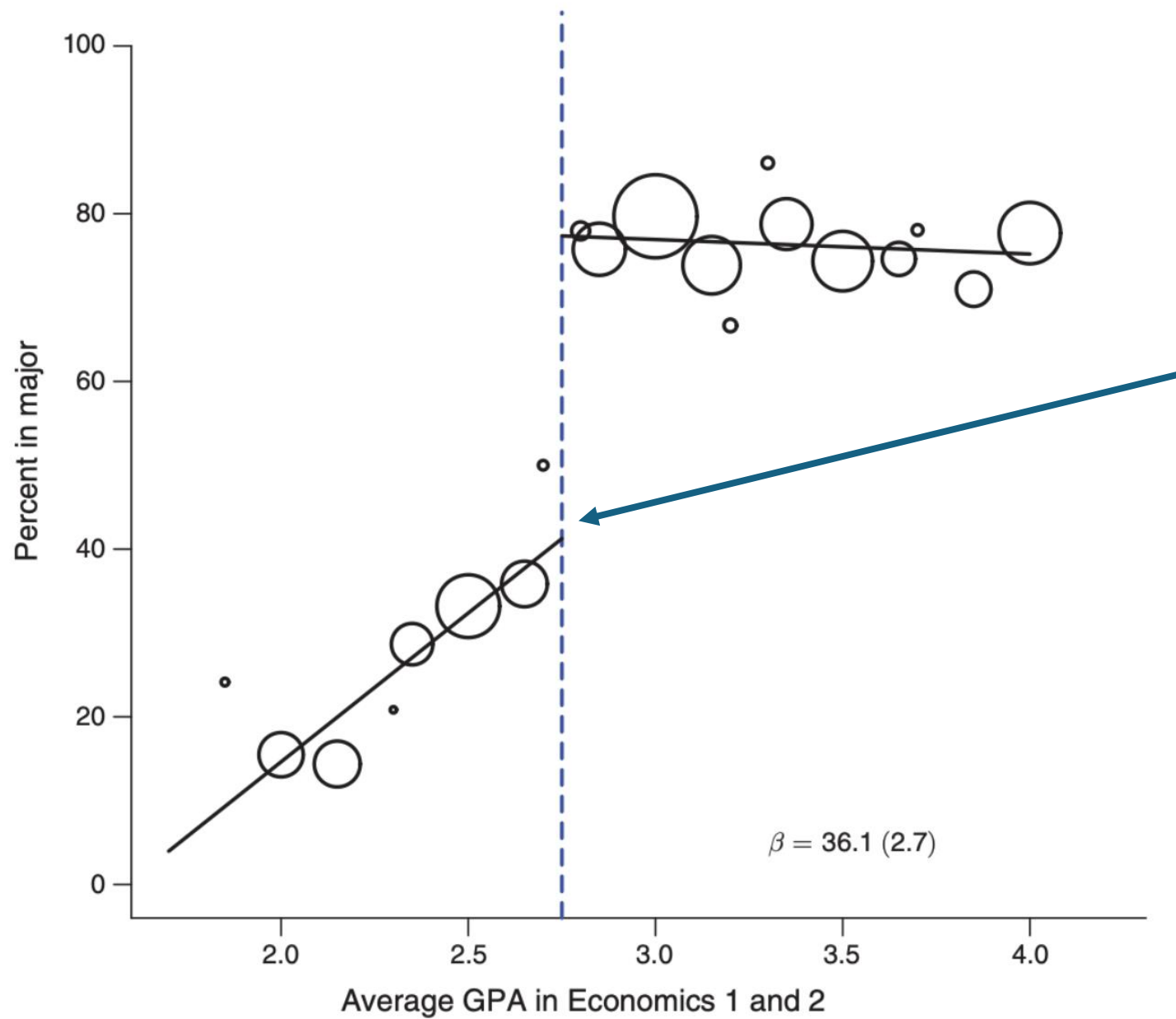
Bleemer and Mehta (2022)

- What is the causal impact of earning an Economics degree?
- Ideal experiment: randomize students into college majors and see how earnings vary across the majors
- Of course this is impossible. But Bleemer and Mehta (2022) again take advantage of a regression discontinuity design to estimate the causal impact of earning an Economics degree



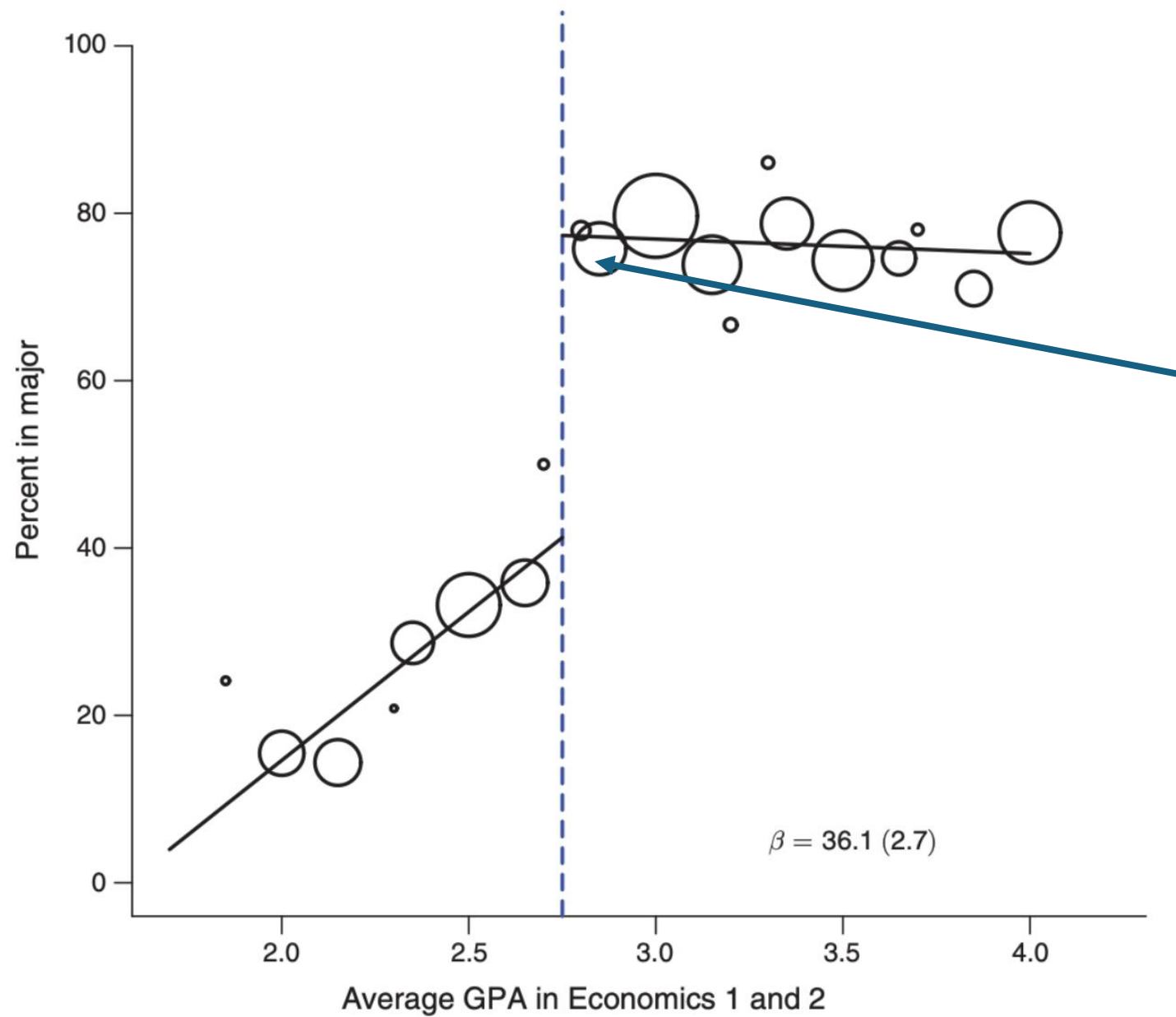
At UC Santa Cruz
you are eligible for
econ major if you
get above a 2.8
gpa in Econ 1 and
Econ 2

FIGURE 1. THE EFFECT OF THE UCSC ECONOMICS GPA THRESHOLD ON MAJORING IN ECONOMICS



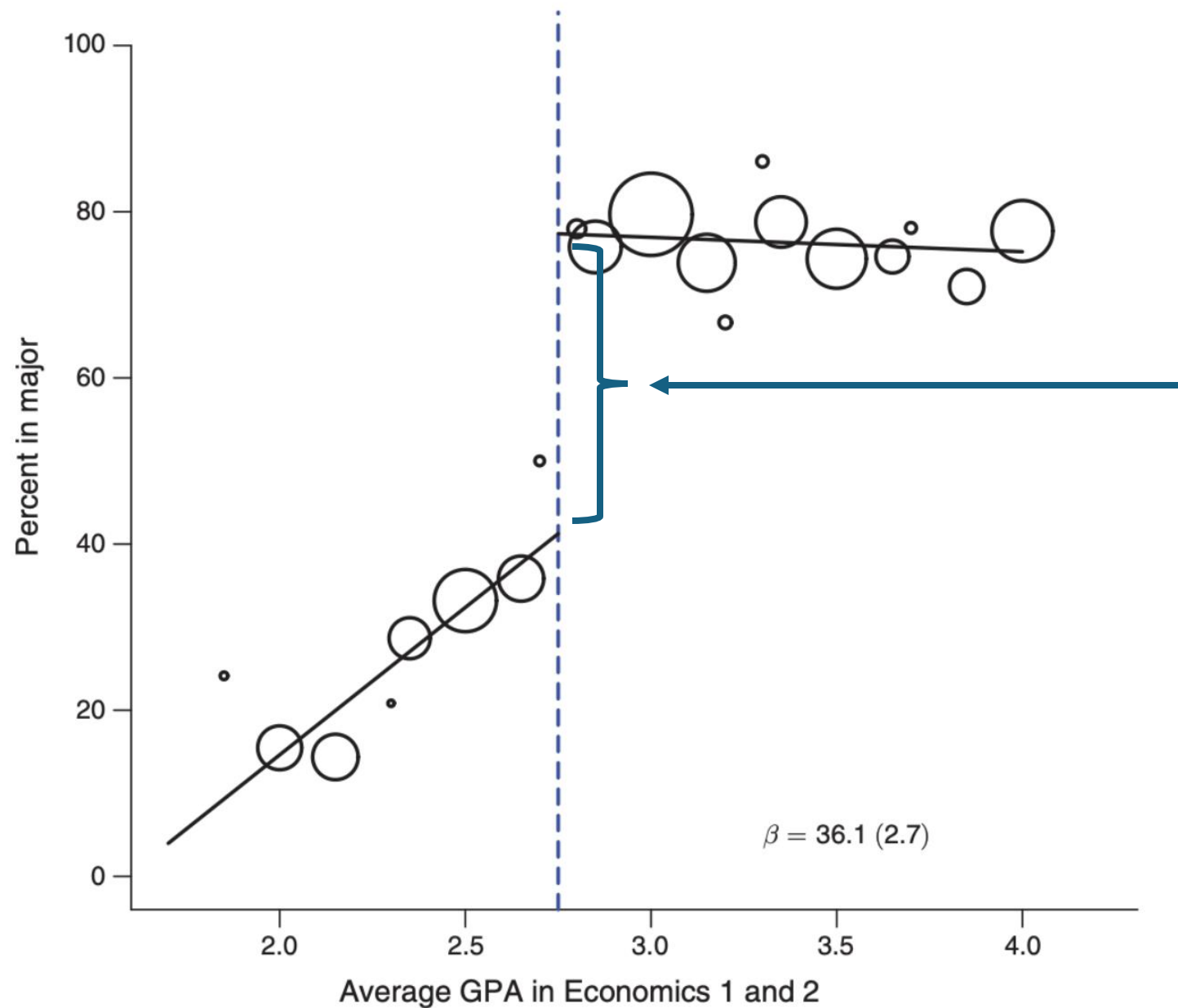
Some get into the program anyways by petitioning

FIGURE 1. THE EFFECT OF THE UCSC ECONOMICS GPA THRESHOLD ON MAJORING IN ECONOMICS



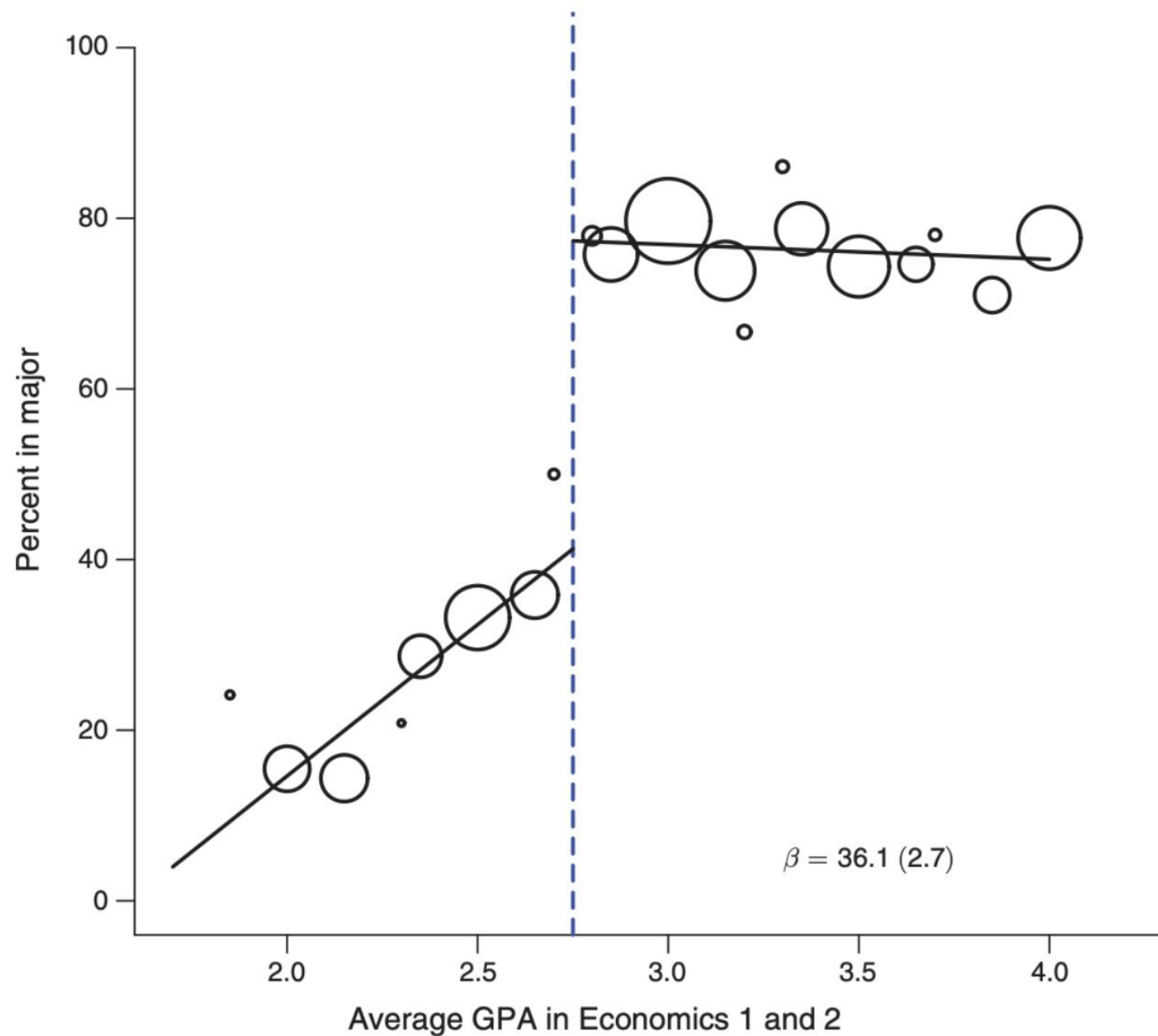
Some are eligible for degree, but choose a different major

FIGURE 1. THE EFFECT OF THE UCSC ECONOMICS GPA THRESHOLD ON MAJORING IN ECONOMICS



Nevertheless,
huge impact of
crossing threshold
on probability of
receiving an
Economics degree

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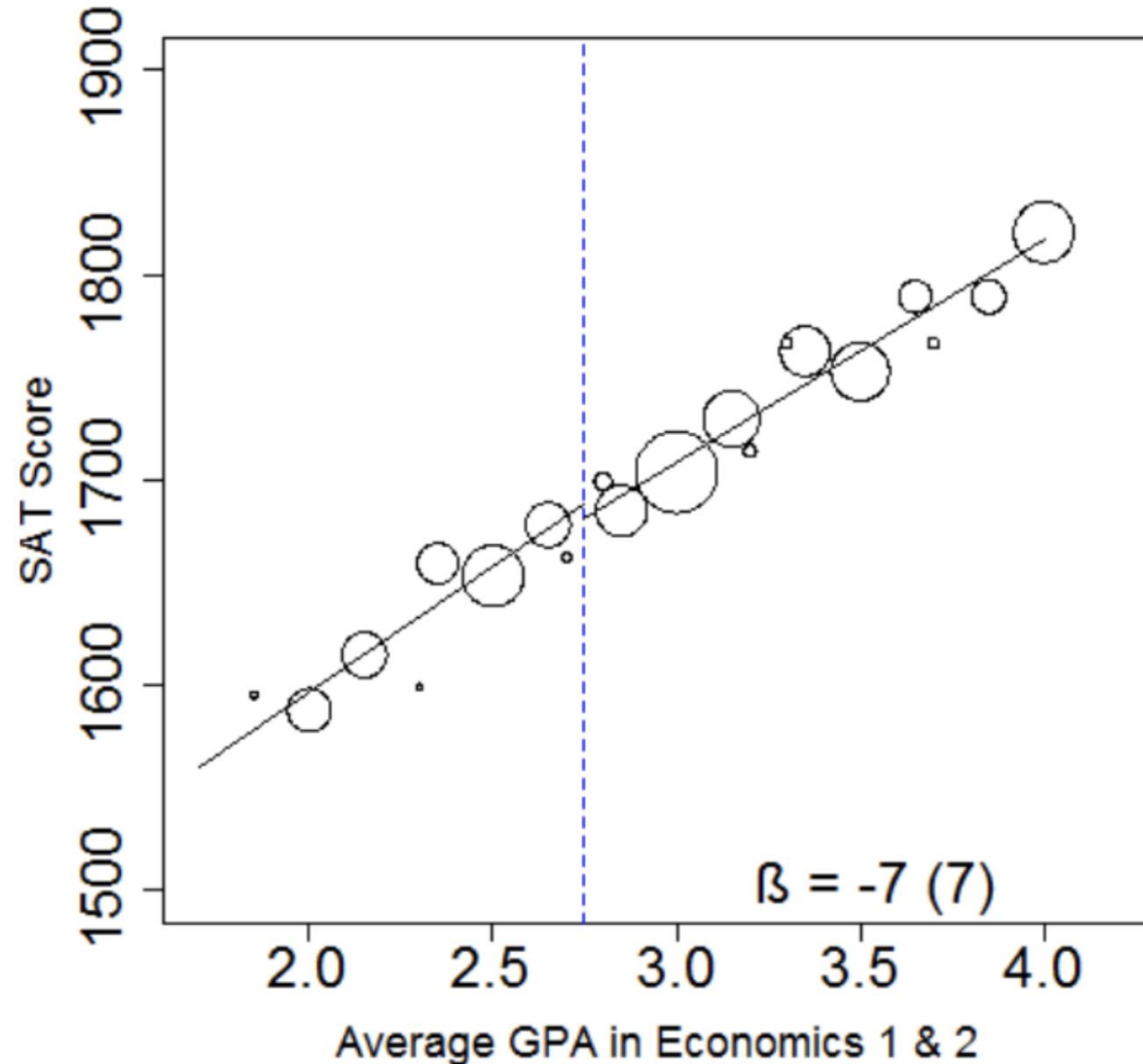


But maybe people to the left are very different to the people on the right of the cutoff

Bleemer and Mehta (2022) can directly test for this using SAT scores

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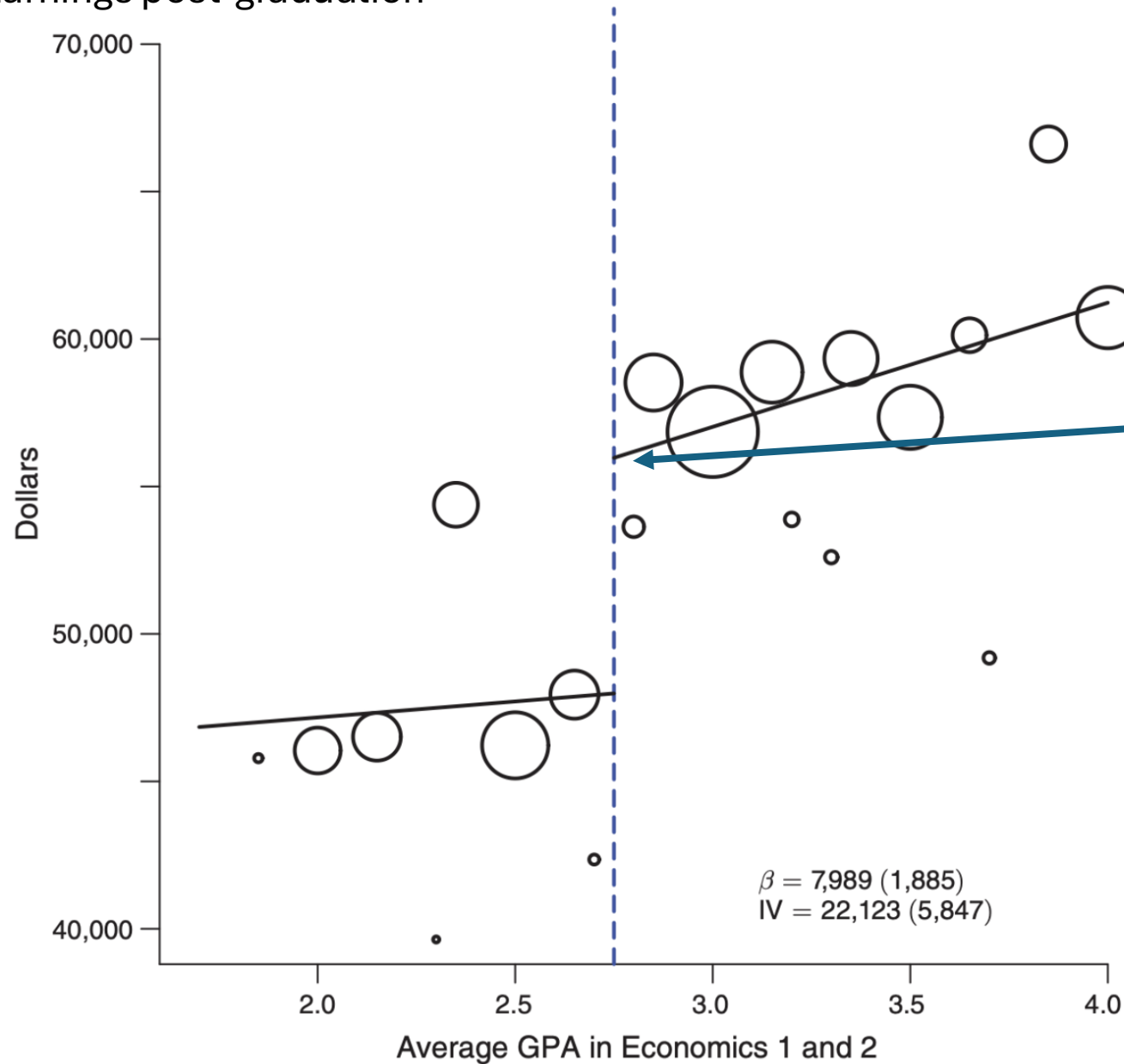
(d) SAT Score



But maybe people to the left are very different to the people on the right of the cutoff

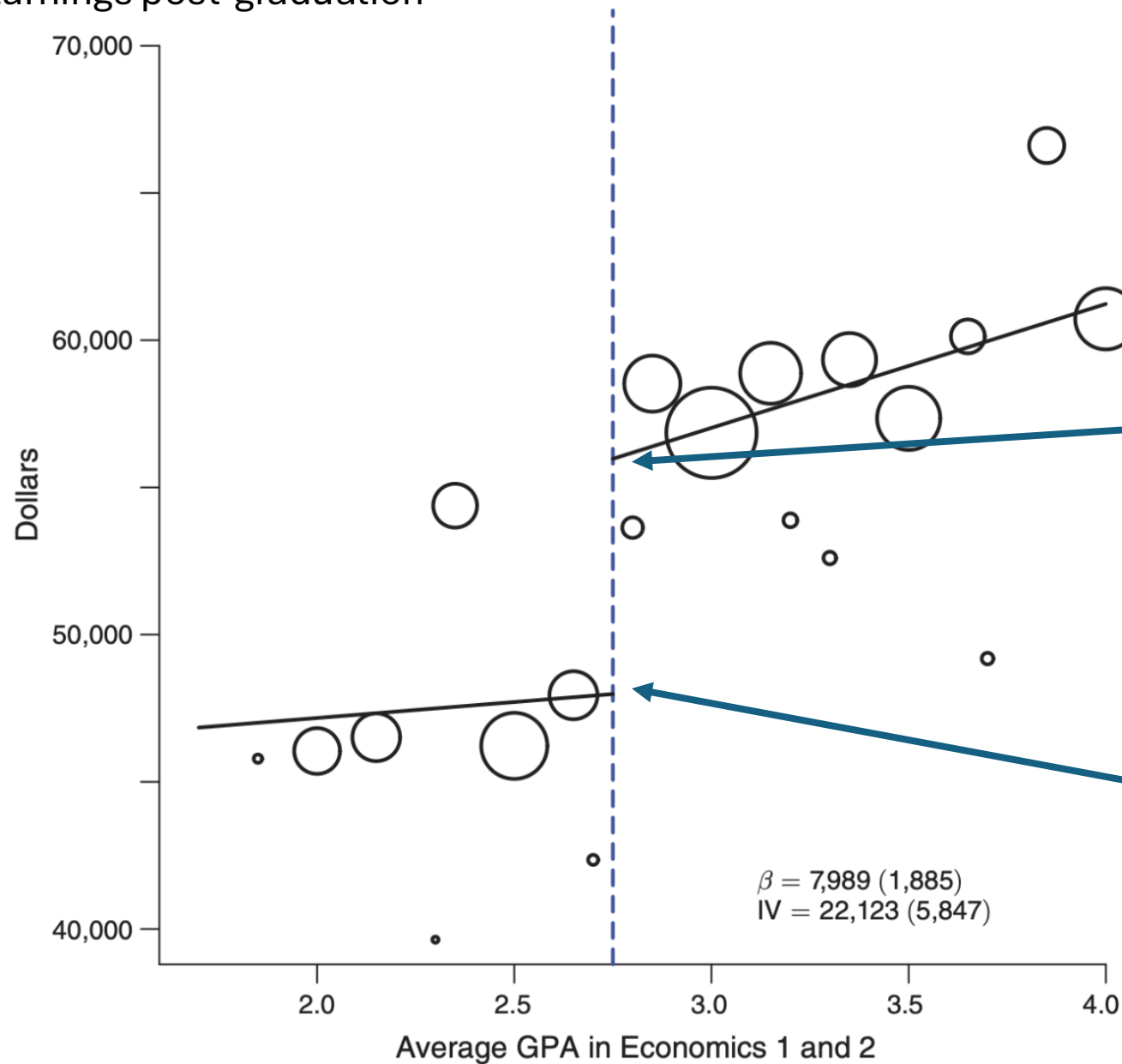
Bleemer and Mehta (2022) can directly test for this using SAT scores

Earnings post-graduation



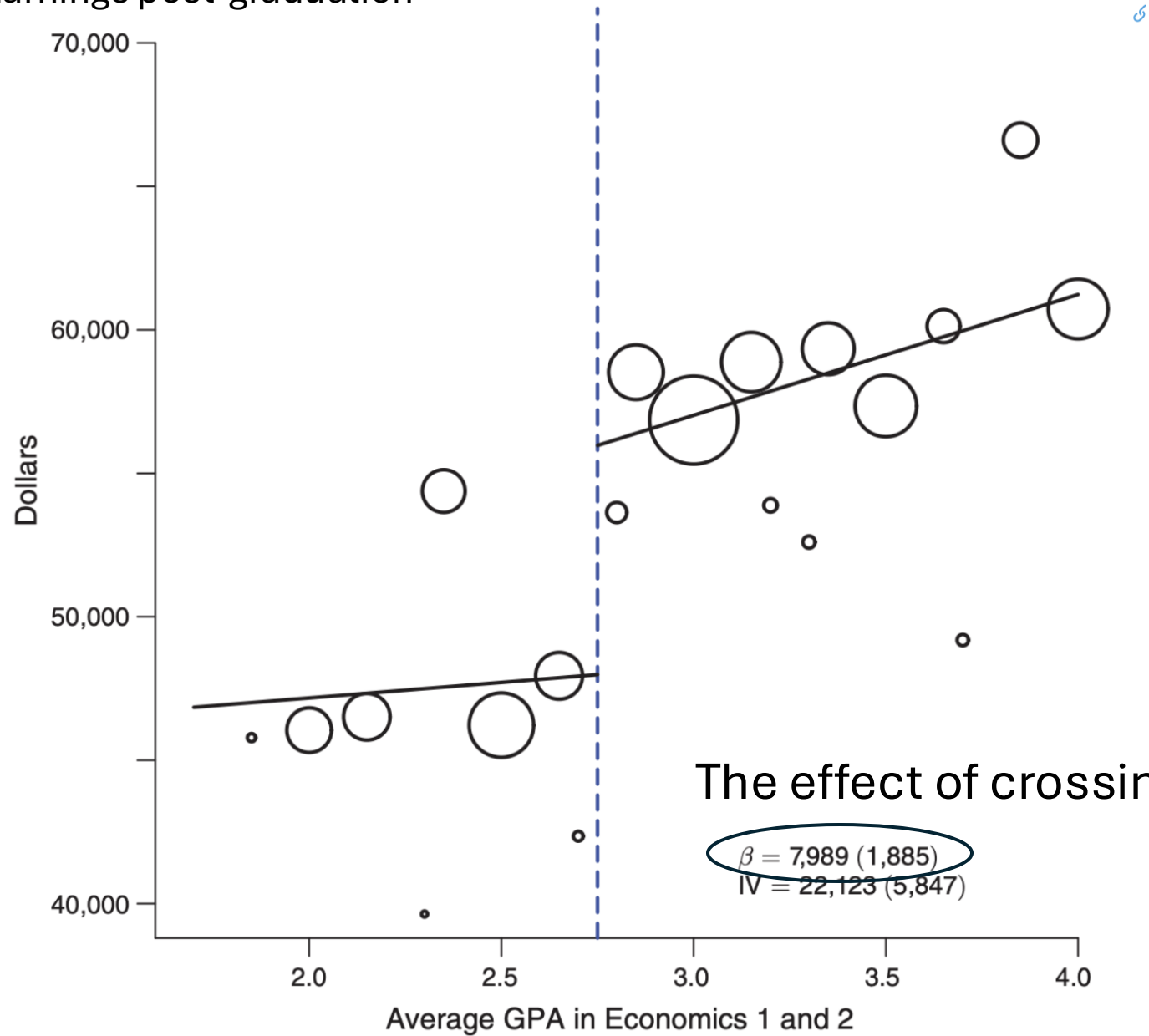
- Those above cutoff earn about 55,000 per year (shortly after graduating)

Earnings post-graduation



- Those above cutoff earn about 55,000 per year (shortly after graduating)
- Those below earn about 47,000

Earnings post-graduation

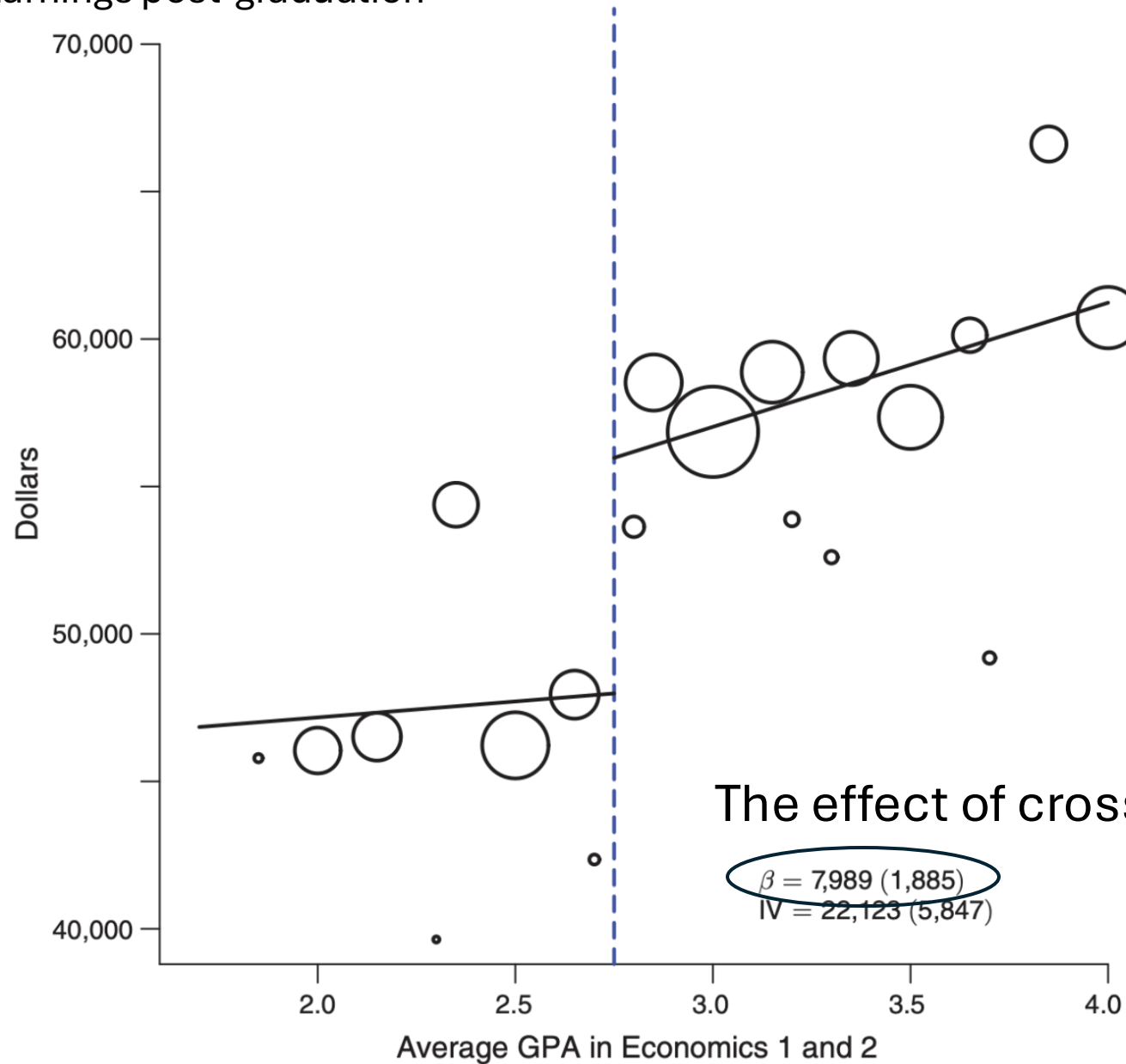


The effect of crossing the threshold on earnings

$$\beta = 7,989 (1,885)$$

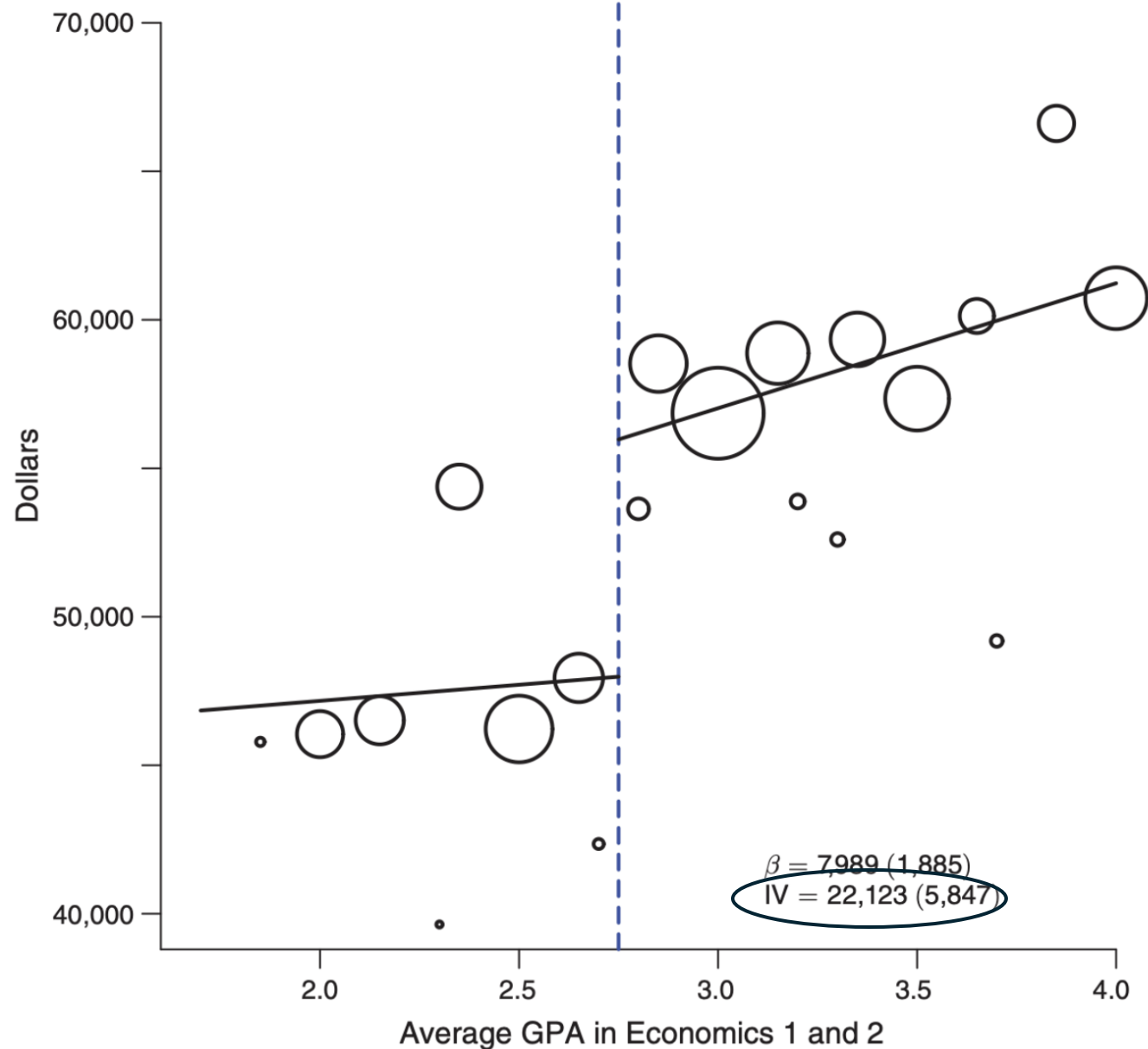
$$IV = 22,123 (5,847)$$

Earnings post-graduation



Scaling this effect by the increase in the probability of receiving an economics degree gets you the causal impact of an economics degree on earnings

Earnings post-graduation



An economics degree increases your earnings by about 22,000 relative to next best option

Scaling this effect by the increase in the probability of receiving an economics degree gets you the causal impact of an economics degree on earnings

Summary



Education increases human capital, causing both increases in your own income as well as increases in aggregate productivity in the economy



Estimating returns to education can be tricky though – individuals who pursue higher education may be different from those who don't



Many clever designs, however, have validated large returns to education